

Vincent Chen

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SUMMARY

Researcher and student leader with hands-on experience in software and GIS development, large-scale program management, and synthetic and materials chemistry. Nationally recognized for excellence in mathematics, computer science, and chemistry.

EDUCATION

Massachusetts Institute of Technology (MIT), Cambridge — *Class of 2030* (Aug 2026 – May 2030)

- **Bachelor of Science (S.B.)** degree in Course 18 (Mathematics) + Course 6-3 (Computer Science and Engineering).
- **GPA:** 5.0/5.0.

Stuyvesant High School, New York — *Class of 2026* (Sep 2022 – Jun 2026)

- **GPA:** 98/100 | **SAT:** 1580 (800 M, 780 EBRW) | **PSAT:** 1520 (760 M, 760 EBRW)
 - Awarded New York State Scholarship for Academic Excellence (top 1% GPA)
- **Relevant Coursework:** Multivariable Calculus, Cybersecurity, Systems-Level Programming, AP Physics C, Organic Chemistry I/II, Advanced Chemistry Lab, Math Team. **Self-Studied:** Abstract Algebra, Linear Algebra, Inorganic + Physical Chemistry.
- **Extracurriculars:** President of Computer Science Dojo, Math Survey, Stuyvesant Alumni Mentoring Program, ACS Chemistry.

WORK HISTORY

NYC Dept. of Parks & Recreation — *Management & Operations Intern* (Jun 2024 – Aug 2026)

- Led projects worth **\$70,000+** and managed **700+** volunteers over 1,200+ man hours in Lower Manhattan parks.
- Developed a horticulture webapp using ArcGIS Pro & Python, mapping **300+** work sites in Manhattan's 200+ parks, streamlining project tracking and aiding data-driven planning for NYC Parks staff and corporate sponsors.
- Honed skills in leadership, GIS, public speaking, and emergency management with 20+ FEMA Certifications in Management/Project Planning; awarded **Gold Presidential Volunteer Service Award**.

American Chemical Society @ Rutgers University Newark — *Research Intern* (Apr 2025 – Aug 2026)

- Synthesized HfO₂-doped carbon nanotube films to investigate ferroelectric properties, with 33-fold increase in H-diffusion.
- Designed/executed experiments involving scanning electron microscopy (SEM), Energy dispersive X-ray spectroscopy (EDXS), and data analysis via Excel/Python and chronopotentiometry to evaluate catalytic morphology and activity.
- Collaborated with an interdisciplinary team to investigate multifunctional oxide materials with energy-related applications, presenting findings at **ACS Fall 2025 Sci-Mix Conference**.

Summer Science Program — *Synthetic Chemistry Researcher* (Jun 2025 – Jul 2025)

- One of **24 selected from 3700+** students for a 5-week intensive research experience in organic & inorganic synthesis.
- Completed a multi-step synthesis of 6 tetraazamacrocyclic compounds, applying concepts from kinetics, reaction mechanisms, and retrosynthesis, with emphasis on oxidation catalysts for environmental remediation (dye bleaching efficacy).
- Collaborated with peers and faculty to write a formal research report and poster, with emphasis on characterization and results, and delivered a conference-style presentation.

AWARDS

- **2026 Regeneron STS Scholar:** Ranked top 300/2,612 applications; awarded \$2,000 for research on HfO₂@CNT membranes.
- **4x AIME Qualifier:** Ranked top 5% in olympiad math, qualified 4x for American Invitational Mathematics Examination (AIME).
- **ACS Catalyst Scholar:** Received a \$40,000 scholarship toward undergraduate studies in science/engineering.
- **2x USAMTS Winner:** Awarded twice by the USA Mathematical Talent Search, a proof-based contest sponsored by the NSA.
- **USACO Silver:** Advanced to Silver (15%) in USA Computing Olympiad.

SKILLS

Languages: Python, Java, C/C++, HTML/CSS.

Concepts: Algorithms, Data Structures, OOP, Cryptography, Memory Management, Networking, Generative AI, Linux, MacOS.